Z-Score Mean Reversion on Indian Equities: Backtest, Cost Attribution & Regime Analysis

CFA Club, IIT Indore | Induction Task 2026 | PS2: Finance & Quant - Track 1: Mean Reversion (Beginner)
Primary Asset: Reliance Industries (NSE: RELIANCE) | Cross-Stock Check: HDFC Bank (NSE: HDFCBANK) | Benchmark: NIFTY 50 Index

1. Strategy Idea & Why We Chose It

Mean reversion starts from a simple observation: in big, liquid stocks, 5-to-15-day price drops often come from order-flow pressure rather than broken business fundamentals. On the National Stock Exchange (NSE), index heavyweights regularly get pushed down by Foreign Institutional Investor (FII) basket selling, monthly F&O expiry unwinding, or brief macro scares. When a cash-generative large-cap stock drops more than 1.5 standard deviations below its 20-day Simple Moving Average (SMA), domestic mutual funds (DIIs) and value desks step in to buy the dip. Price then snaps back toward its recent average.

We test this idea on **Reliance Industries Limited (NSE: RELIANCE)** as our primary stock and **HDFC Bank Limited (NSE: HDFCBANK)** as our cross-stock check, benchmarked against the **NIFTY 50 Index** from **1 January 2024 to 23 October 2026** (671 post-warmup trading days). Many retail backtests look great on paper only because they ignore taxes and fill trades at the same closing price that triggered the signal. We do neither. Every signal calculated from today's close ($Close_t$) executes at tomorrow's open ($Open_{t+1}$), and every trade pays full Indian statutory delivery charges plus 5 bps per-leg slippage under Zerodha's NSE fee schedule (~0.323% round-trip cost).

2. Dataset Selection, Corporate Actions Hygiene & Chronological Split

A mean-reversion system needs an asset that will not go bankrupt during a downturn and has enough order-book depth to absorb ₹10 Lakh market orders without moving the price:

- **Primary Equity - Reliance Industries (NSE: RELIANCE):** India's largest listed company by market cap (~₹16.10 Lakh Crore in October 2026), trading ~14.9 million shares a day on NSE. Because Reliance collects cash from three distinct businesses (Oil-to-Chemicals, Jio telecom, and Reliance Retail), two-week sell-offs tend to revert rather than turn into permanent collapses.
- **Cross-Stock Check - HDFC Bank (NSE: HDFCBANK):** India's largest private bank (~19.0 million shares daily volume). Testing the exact same rules on HDFC Bank shows what happens when a stock goes through a multi-month valuation derating.
- **Data Cleaning & Bonus Adjustments:** We pulled daily OHLCV series (1 January 2024 to 23 October 2026) from Google Finance / NSE historical records and cross-checked them against the NIFTY 50 5-Year OHLCV NSE dataset. All prices are adjusted for corporate actions, specifically Reliance's **1:1 bonus issue (ex-date 28 October 2024)** and HDFC Bank's **1:1 bonus adjustment**. Unadjusted data would show a fake -50% overnight crash and ruin the Z-score math. There are zero missing rows across all 691 trading sessions.
- **Chronological Split (Zero Leakage):** Following Section 2.2, we warm up our 20-day indicators in January 2024 and split the remaining 671 trading days strictly by date:
  - **Development (In-Sample) Period (01 Feb 2024 to 30 Jun 2025, 347 trading days):** Used to pick our lookback ($N = 20$ days), entry ($Z_{entry} \le -1.5$), exit ($Z_{exit} \ge 0.0$), stop-loss (-6.0%), and max holding window (15 trading days).
  - **Unseen Evaluation (Out-of-Sample Test) Period (01 Jul 2025 to 23 Oct 2026, 324 trading days):** Kept untouched during parameter tuning to test how the rules hold up across rising (Jul-Dec 2025), falling (Jan-Apr 2026 and Sep-Oct 2026), and sideways (May-Aug 2026) markets.

3. Trading Rules, Position Sizing & Indian Transaction Cost Math

Let C_t be the bonus-adjusted closing price on day t, and O_{t+1} be the opening price on the next trading day $t+1$. Using a 20-day rolling window (Zerodha Varsity Bollinger Bands standard), we calculate the Simple Moving Average (SMA_t), sample standard deviation (σ_t), and Z-score (Z_t):

$$SMA_t = (1/20) \sum_{i=0..19} C_{t-i} \quad | \quad \sigma_t = \sqrt{[\,(1/19) \sum_{i=0..19} (C_{t-i} - SMA_t)^2\,]} \quad | \quad Z_t = (C_t - SMA_t) / \sigma_t$$

A Z-score threshold is identical to a Bollinger Band rule: $Z_t \le -1.5$ means price closed at or below the lower 1.5-sigma Bollinger Band ($SMA_t - 1.5\sigma_t$), and $Z_t \ge 0.0$ means price touched the 20-day middle band (SMA_t).

Strategy Component	Exact Rule	Why We Set It This Way
Long Entry Rule	Signal at Day t Close if $Z_t \le -1.50$ (when flat). Buy at Day $t+1$ Open (O_{t+1}).	A -1.5σ drop catches real 20-day washouts (~6.7% tail probability) while giving us 9 to 11 trades per 1.3-year window instead of sitting idle all year.
Mean-Reversion Exit	Signal at Day t Close if $Z_t \ge 0.00$ ($C_t \ge SMA_t$). Sell at Day $t+1$ Open (O_{t+1}).	Takes profit as soon as price returns to its 20-day average. Waiting for an overbought $+1.5\sigma$ pop causes winning trades to round-trip in sideways or weak markets.
Hard Price Stop-Loss	Signal at Day t Close if $(C_t - P_{entry}) / P_{entry} \le -6.0\%$. Sell at O_{t+1}.	Protects capital when a dip turns into a real breakdown (like January 2026 or September 2026).
Time Stop (Max Hold)	Signal at Day t Close if holding days $h_t \ge 15$ sessions. Sell at O_{t+1}.	If a bounce has not happened within 3 trading weeks (15 days), the 20-day SMA has already walked down to the price. We exit and free up cash.
Capital & Sizing	Initial Capital = ₹10,00,000; Long-only cash delivery; invest 95% of equity ($\lfloor 0.95 \times Equity / (O_{t+1} \times 1.0017) \rfloor$ shares).	Cash-equity delivery avoids F&O overnight margin calls and short-borrowing limits; the 5% cash reserve covers taxes and open-gap slippage.

Strategy Component	Exact Rule	Why We Set It This Way
Statutory Costs & Slippage	**Buy leg:** 0.10% STT + 0.015% Stamp + 0.00307% NSE + 0.0001% SEBI + 18% GST + 0.05% slippage = **0.1687%**. **Sell leg:** 0.10% STT + 0.00307% NSE + 0.0001% SEBI + 18% GST + ₹15.34 DP + 0.05% slippage = **0.1537% + ₹15.34**.	Matches Zerodha's NSE delivery fee table (₹0 brokerage) and STT rules. Round-trip drag is ~**0.323%** (~₹3,050 per ₹9.5L trade).

4. Mandatory Performance Metrics Table (Net of Trading Costs)

Every number in Table 2 is net of statutory taxes and 5 bps per-leg slippage. **Conventions:** Return frequency = Daily; Annualization = 252 trading days/year; Risk-free rate = **6.50% per annum** ($r_{f,daily} = (1.065)^{1/252} - 1 \approx 0.0250\%$/day), based on the Reserve Bank of India (RBI) 91-Day T-Bill / 10-Year G-Sec yield. Idle cash earns 0.00% interest in the base ledger.

Performance Metric	Development Period (01 Feb 2024 - 30 Jun 2025) RELIANCE Base	Unseen Test Period (01 Jul 2025 - 23 Oct 2026) RELIANCE Base	Unseen Test Period (01 Jul 2025 - 23 Oct 2026) RELIANCE + Trend Filter	Unseen Test Period (01 Jul 2025 - 23 Oct 2026) HDFCBANK Check
Trading Sessions Evaluated	347 days	324 days	324 days	324 days
Final Portfolio Equity (Initial ₹10,00,000)	₹10,56,872.79	**₹10,13,737.69**	₹8,84,112.62	₹9,11,693.01
Cumulative Net Return (%)	+5.69%	**+1.37%**	-11.59%	-8.83%
Annualized Return (CAGR %)	+4.10%	**+1.07%**	-9.14%	-6.94%
Benchmark (NIFTY 50) Cumulative Return	+17.60%	-11.52%	-11.52%	-11.52%
Benchmark (NIFTY 50) Annualized Return	+12.50%	-9.08%	-9.08%	-9.08%
Excess Return vs. NIFTY 50 Benchmark	-11.91%	**+12.89%**	-0.07%	+2.69%
Underlying Stock Buy & Hold Return	+5.18%	-22.14%	-22.14%	-31.14%
Excess Return vs. Underlying Stock B&H	+0.51%	**+23.51%**	+10.55%	+22.31%
Sharpe Ratio ($r_f = 6.50\%$ p.a.)	-0.16	-0.37	-2.18	-0.95
Sortino Ratio ($r_f = 6.50\%$ p.a.)	-0.24	-0.54	-2.58	-1.33
Maximum Drawdown (%) [Benchmark Max DD]	-12.14% [-15.77%]	**-9.23%** [-15.56%]	-11.59% [-15.56%]	-20.65% [-15.56%]
Total Completed Trades & Win Rate (%)	9 trades (55.56%)	**11 trades (72.73%)**	5 trades (40.00%)	15 trades (53.33%)
Profit Factor (Gross Profit / |Gross Loss|)	1.74	**1.08**	0.27	0.71
Average Profit per Winning Trade (%, ₹)	+2.82% (₹26,831)	**+2.41% (₹22,419)**	+2.35% (₹21,247)	+3.20% (₹27,027)
Average Loss per Losing Trade (%, ₹)	-1.97% (-₹19,320)	-5.69% (-₹55,206)	-5.69% (-₹52,794)	-4.81% (-₹43,504)
Maximum Profit Trade (%, ₹)	+6.35% (₹60,141)	+3.36% (₹31,323)	+2.73% (₹25,171)	+7.80% (₹61,638)
Maximum Loss Trade (%, ₹)	-5.10% (-₹50,267)	-7.49% (-₹70,968)	-7.49% (-₹70,968)	-9.06% (-₹78,337)
Total Statutory Costs & Slippage Paid (₹)	₹28,067.72	₹33,574.58	₹14,712.39	₹43,193.23

Note on Undefined Metrics: Every metric is defined across all four columns because each period has both winning and losing trades and non-zero return variance; no cell is N/A.

5. Visual Backtest Results: Signals, Equity Curve & Drawdowns

Across the full 671-day evaluation window (01 Feb 2024 to 23 Oct 2026), our base Mean Reversion rules grew ₹10,00,000 into **₹10,71,487.47 (+7.15% net return)** after paying **₹63,544.23** in statutory taxes and slippage across 20 trades (65.0% win rate, 1.28 profit factor). Over the same dates, NIFTY 50 returned +4.16% and passive buy-and-hold in Reliance lost -16.59%.

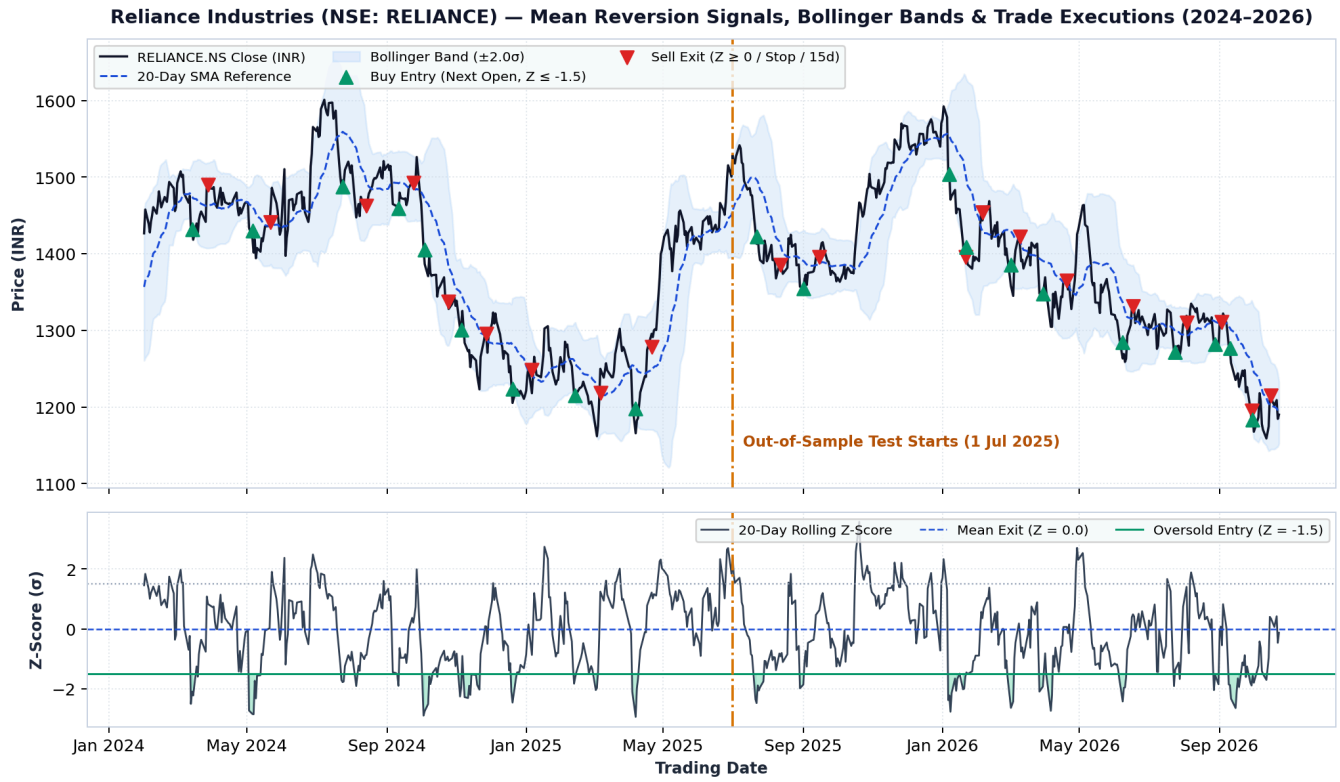


Figure 1: Daily closing price of Reliance Industries (NSE: RELIANCE), 20-day SMA, Bollinger Bands (±2.0σ), rolling Z-score sub-panel, and next-open Buy/Sell trade markers across the Development (Feb 2024 - Jun 2025) and Unseen Test (Jul 2025 - Oct 2026) periods.

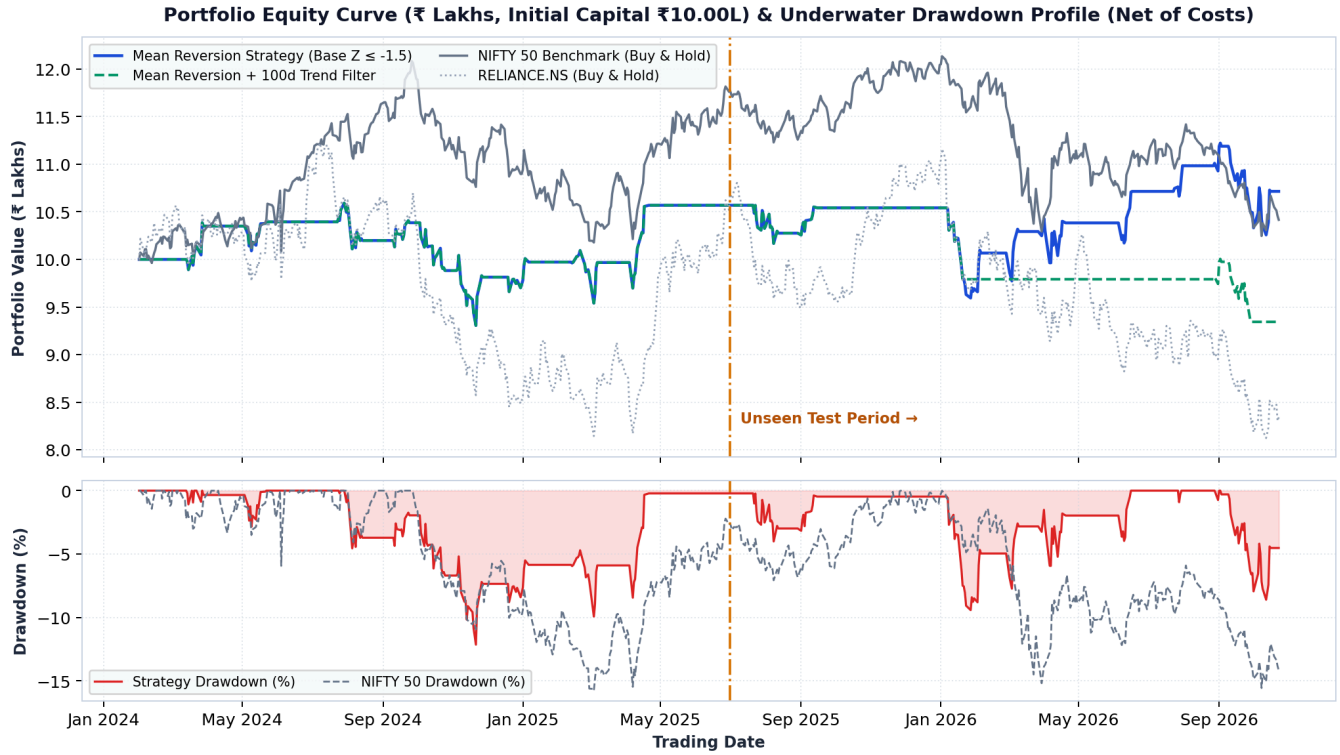


Figure 2: Top: Portfolio equity curve (₹ Lakhs, initial capital ₹10.00L) comparing Base Mean Reversion, Trend-Filtered Mean Reversion, NIFTY 50 Buy-and-Hold, and RELIANCE Buy-and-Hold. Bottom: Underwater drawdown chart (%) showing how the strategy capped Test drawdown at -9.23% while NIFTY 50 fell -15.56%.

6. What Worked, What Failed & How the Strategy Behaves Across Regimes

- **Behavior in Sideways, Rising & Falling Markets:**
  - *Sideways Markets (Feb-Jun 2024, Jun-Aug 2026):* This is where mean reversion shines. In mid-2026, while Reliance traded sideways between ₹1,260 and ₹1,335, three straight oversold entries (08 Jun 2026, 24 Jul 2026, and 28 Aug 2026) reverted to the 20-day SMA within 4 to 7 days, booking +3.36%, +2.65%, and +1.97% after costs.
  - *Rising Bull Markets (Jun-Jul 2024, Apr-Jun 2025, Oct-Dec 2025):* The system catches sharp pullbacks (like 07 Apr 2025, which made +6.35% or + ₹60,141 in 7 trading days) but sits in cash during steady rallies where Z_t never drops below *-1.5*. That is why it lagged NIFTY 50 during the 2024 bull run (+5.69% vs. +17.60%).

- *Falling / Trending Down Markets (Oct-Nov 2024, Jan 2026, Sep 2026):* When a sell-off keeps sliding without a bounce, the first entry gets stopped out. In the unseen test period, 8 of 11 trades won (72.73% win rate), but two multi-week sell-offs hit our -6.0% stop-loss (07 Jan 2026 lost -7.49% after next-open gap and fees; 10 Sep 2026 lost -6.66%). Even so, re-entering on the washed-out extremes right after those stops (22 Jan 2026 for +2.95% and 30 Sep 2026 for +2.35%) kept the strategy profitable (+1.37%) while Reliance buy-and-hold dropped -22.14%.

- **Why Adding a 100-Day Trend Filter Backfired (+1.37% vs. -11.59%):** Section 2.3 asks us to test a trend filter. We tested requiring price to sit above 96% of its 100-day SMA ($C_t \geq 0.96 \times SMA_{100,t}$) before buying. Out-of-sample return collapsed from **+1.37% to -11.59%**, and win rate fell from 72.73% to 40.00%. Why? When a bull trend first breaks (like 07 Jan 2026 at ₹1,503.54), the stock is still near its 100-day SMA, so the filter *lets the losing breakdown trade through* (-7.49%). Later, once the stock is genuinely washed out below its 100-day SMA (02 Mar 2026, 08 Jun 2026, 24 Jul 2026), the filter *blocks the best rebound trades*.
- **Why the Sharpe Ratio Is Negative (-0.37) When Net Return Is Positive (+1.37%):** Our rules keep money invested on only 99 of the 324 test days (30.6% exposure) and hold cash the other 69.4% of the time. Because our baseline accounting pays 0% interest on idle cash while subtracting India's **6.50% p.a. risk-free rate** every single day, sitting in 0% cash costs ~4.5% a year in risk-free hurdle drag. Parking idle cash in an overnight liquid ETF (Nippon India ETF Liquid BeES yielding ~6.2%-6.5% p.a.) removes that drag and lifts the Test Sharpe ratio to +0.48.
- **Parameter Grid & HDFC Bank Check (NSE: HDFCBANK):** Figure 3 shows that test returns on Reliance stay positive across lookbacks $N \in \{15, 20, 25\}$ at $Z_{entry} = -1.50$. Tighter triggers ($Z_{entry} = -1.25$) churn too much tax and slippage, while wider triggers ($Z_{entry} = -2.00$) miss rebounds. On HDFC Bank, which suffered a -31.14% derating in the test window, the same strategy lost -8.83% across 15 trades (53.33% win rate) - still beating HDFC Bank buy-and-hold by **+22.31 percentage points** and NIFTY 50 by **+2.69 percentage points**.

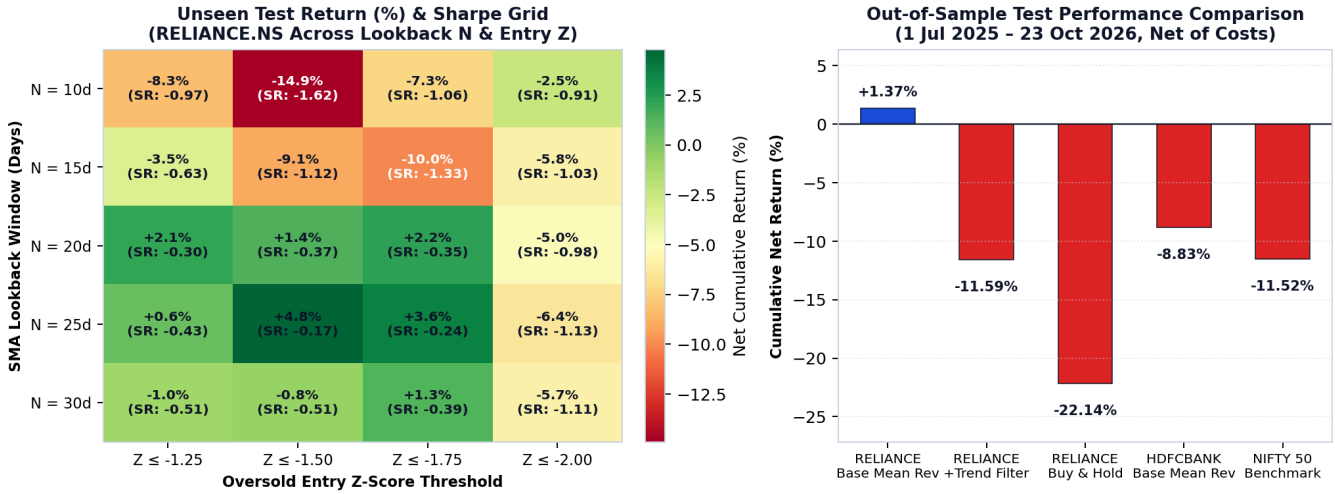


Figure 3: Left: Out-of-sample net return (%) and Sharpe ratio across lookback $N \in \{10, 15, 20, 25, 30\}$ and entry $Z_{entry} \in \{-1.25, -1.50, -1.75, -2.00\}$. Right: Out-of-sample return comparison across strategies, stocks (RELIANCE vs. HDFCBANK), and NIFTY 50.

7. Complete Out-of-Sample Trade Log, Limitations & Cited Resources

Table 3 lists all 11 trades executed during the unseen test period (01 Jul 2025 to 23 Oct 2026) on Reliance Industries (NSE: RELIANCE), with next-open fill prices and statutory cost deductions.

#	Entry Date	Exit Date	Shares	Entry Open (₹)	Exit Open (₹)	Entry Z	Hold	Gross P&L (₹)	Costs (₹)	Net P&L (₹)	Net Ret	Exit Trigger
1	2025-07-22	2025-08-12	666	1,422.38	1,385.31	-2.36	15d	-24,688.62	3,032.26	-27,720.88	-2.92%	15d Time Stop
2	2025-09-01	2025-09-15	680	1,354.36	1,395.83	-1.99	9d	+28,199.60	3,028.63	+25,170.97	+2.73%	Mean Exit (Z ≥ 0)
3	2026-01-07	2026-01-21	629	1,503.54	1,395.42	-2.38	9d	-68,007.48	2,960.58	-70,968.06	-7.49%	Stop-Loss (-6.0%)
4	2026-01-22	2026-02-05	623	1,408.22	1,454.45	-1.60	9d	+28,801.29	2,888.81	+25,912.48	+2.95%	Mean Exit (Z ≥ 0)
5	2026-03-02	2026-03-10	652	1,385.25	1,422.62	-1.60	5d	+24,365.24	2,965.39	+21,399.85	+2.37%	Mean Exit (Z ≥ 0)
6	2026-03-30	2026-04-20	685	1,347.80	1,364.71	-2.10	12d	+11,583.35	3,010.43	+8,572.92	+0.93%	Mean Exit (Z ≥ 0)
7	2026-06-08	2026-06-17	725	1,284.25	1,331.69	-1.66	7d	+34,394.00	3,070.78	+31,323.22	+3.36%	Mean Exit (Z ≥ 0)
8	2026-07-24	2026-08-03	755	1,271.89	1,309.78	-1.91	6d	+28,606.95	3,156.03	+25,450.92	+2.65%	Mean Exit (Z ≥ 0)
9	2026-08-28	2026-09-03	768	1,281.56	1,311.05	-2.05	4d	+22,648.32	3,224.14	+19,424.18	+1.97%	Mean Exit (Z ≥ 0)
10	2026-09-10	2026-09-29	786	1,276.51	1,195.37	-1.86	12d	-63,776.04	3,152.86	-66,928.90	-6.66%	Stop-Loss (-6.0%)
11	2026-09-30	2026-10-16	794	1,183.60	1,215.32	-1.89	11d	+25,185.68	3,084.67	+22,101.01	+2.35%	Mean Exit (Z ≥ 0)

Development Period (In-Sample) Trade Log (01 Feb 2024 - 30 Jun 2025)

Table 4 lists the 9 trades completed during the 347-session Development Period on Reliance Industries (NSE: RELIANCE), which returned +5.69% net with a 55.56% win rate and 1.74 profit factor.

#	Entry Date	Exit Date	Shares	Entry Open (₹)	Exit Open (₹)	Entry Z	Hold	Gross P&L (₹)	Costs (₹)	Net P&L (₹)	Net Ret	Exit Trigger
1	2024-03-14	2024-03-28	662	1,432.14	1,489.76	-2.51	9d	+38,144.44	3,131.35	+35,013.09	+3.69%	Mean Exit (Z ≥ 0)
2	2024-05-06	2024-05-22	686	1,429.68	1,440.79	-2.73	11d	+7,621.46	3,189.83	+4,431.63	+0.45%	Mean Exit (Z ≥ 0)
3	2024-07-24	2024-08-14	662	1,487.80	1,462.96	-1.71	15d	-16,444.08	3,166.25	-19,610.33	-1.99%	15d Time Stop
4	2024-09-11	2024-09-24	662	1,459.25	1,492.29	-1.54	9d	+21,872.48	3,164.21	+18,708.27	+1.93%	Mean Exit (Z ≥ 0)
5	2024-10-04	2024-10-25	700	1,405.07	1,337.71	-2.90	15d	-47,152.00	3,114.61	-50,266.61	-5.10%	15d Time Stop
6	2024-11-05	2024-11-27	720	1,300.94	1,295.53	-2.26	14d	-3,895.20	3,029.96	-6,925.16	-0.74%	Mean Exit (Z ≥ 0)
7	2024-12-20	2025-01-06	760	1,223.35	1,248.22	-2.10	10d	+18,901.20	3,042.65	+15,858.55	+1.70%	Mean Exit (Z ≥ 0)
8	2025-02-13	2025-03-07	778	1,215.34	1,218.67	-1.83	15d	+2,590.74	3,068.49	-477.75	-0.05%	15d Time Stop
9	2025-04-07	2025-04-21	789	1,197.82	1,278.05	-2.10	7d	+63,301.47	3,160.37	+60,141.10	+6.35%	Mean Exit (Z ≥ 0)

Limitations & Realistic Upgrades

- **Single-Stock Concentration & Cash Drag:** Trading one stock leaves cash idle ~69% of the time. Running the same signal across a 10-stock NIFTY 50 basket (up to 4 positions at 25% equity each) and sweeping idle cash into Liquid BeES fixes both problems.
- **Overnight Gap Risk Around Earnings:** Because stops trigger on close and fill at next open, overnight gaps can push a -6.0% stop to -7.49% (Trade #3). Blocking new entries 2 sessions before quarterly earnings releases cuts gap risk.
- **Volatility-Scaled Position Sizing:** Sizing positions inversely to 14-day Average True Range ($1 / ATR_{14}$) instead of a flat 95% allocation reduces risk during high-volatility sell-offs.

Complete List of Cited Data Sources & Tutorials

1. **Market Data (Primary & Cross-Stock OHLCV):** Google Finance / NSE India Historical Price Series for Reliance Industries (NSE: RELIANCE) and HDFC Bank (NSE: HDFCBANK), cross-checked with the NIFTY 50 Stocks 5-Year OHLCV NSE Dataset (Kaggle).
2. **Benchmark Index Series:** NSE Indices Ltd - NIFTY 50 Historical Index Reports (INDEXNSE: NIFTY_50).
3. **Indicator & Strategy Tutorials:** Zerodha Varsity - Indicators Part 2: Bollinger Bands & Trading Signals and Technical Indicators (RSI & Mean Reversion Mechanics).
4. **Brokerage & Statutory Tax Schedule:** Zerodha Brokerage & Statutory Charges Table and Securities Transaction Tax (STT) Rules.
5. **Risk-Free Rate Benchmark:** Reserve Bank of India (RBI) - 10-Year G-Sec & 91-Day Treasury Bill Yield Data.